

LAB 3 – COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Faculty Research Grant & Publication Tracker

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1. Architecture Selection

We chose Microservices Architecture for the Faculty Research Grant & Publication Tracker. The application is divided into independently deployable services for user access, grant management, expenses, publications, approvals, analytics, and auditing. Services communicate through defined interfaces and share persistent information through the research tracker data layer.

2. Architecture Comparison

Architecture	Structure	Advantages	Decision
Layered	Presentation, business and data layers	Simple separation of concerns; easy to understand	Not selected
Client-Server	Centralized server serving clients	Centralized control and simpler deployment	Not selected
Microservices	Independent services communicating through APIs	Independent scaling, fault isolation and independent deployment	Selected

3. Component Diagram

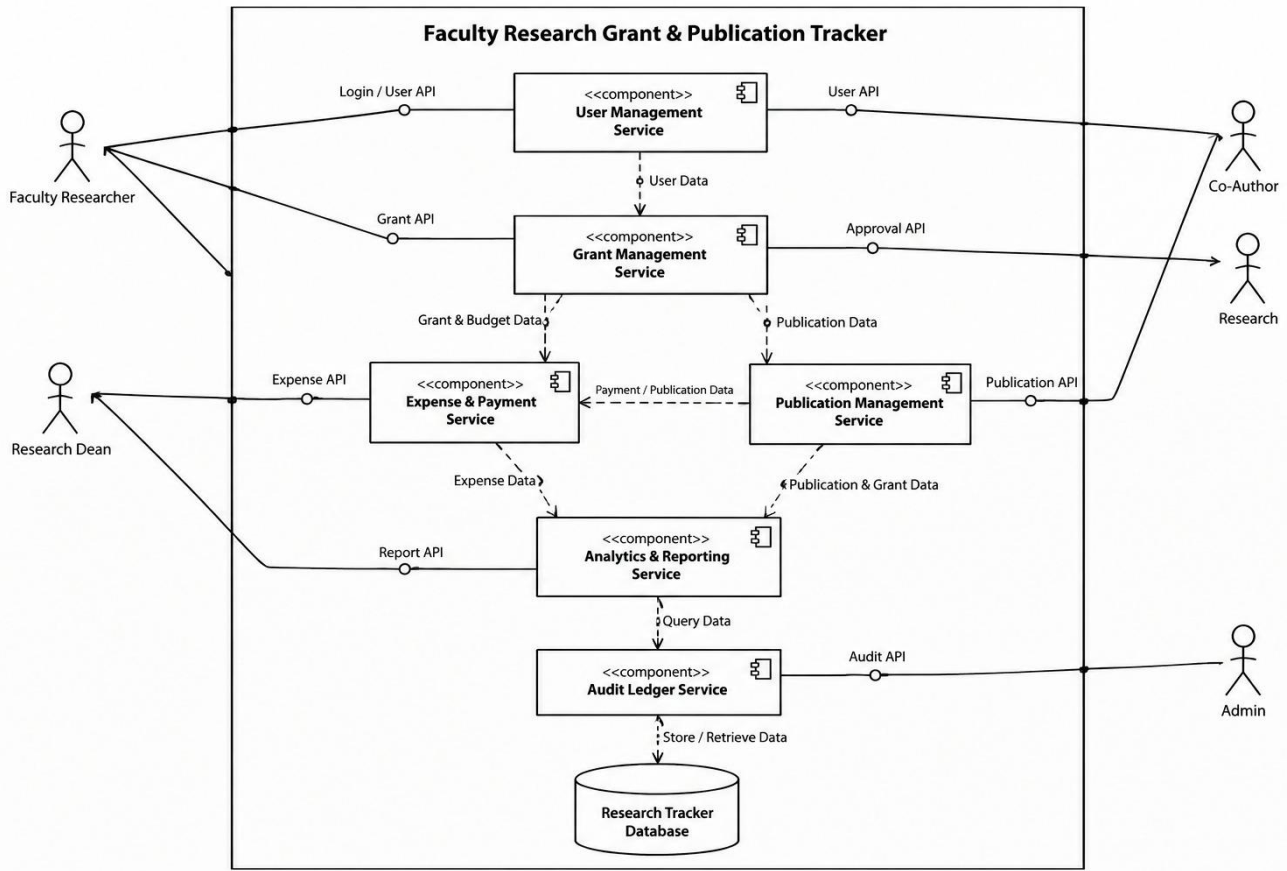


Figure 1: UML Component Diagram

4. Components and Responsibilities

User & Access Management Service: Handles authentication, authorization and role-based access for Faculty Researchers and Research Dean.

Grant Management Service: Maintains grant allocations, expenditure limits and remaining budget balances.

Expense Management Service: Records equipment procurement and other grant expenses and validates available budget.

Publication Management Service: Stores publication details including indexing and citation information and supports co-author approval.

Approval & Workflow Service: Manages Research Dean approvals/rejections and co-author approval workflows.

Analytics & Reporting Service: Generates fund burn-up analytics showing allocation, expenditure and remaining balance.

Audit Ledger Service: Maintains tamper-evident records of financial approvals and journal/status modifications.

Research Tracker Database: Provides persistent storage for grants, expenses, publications, users, approvals and reporting data.

5. Interfaces and Interactions

User/Auth API: Connects User & Access Management with grant and other protected services.

Grant API: Provides grant and budget information to dependent services.

Publication API: Supports publication-related service communication.

Approval API: Transfers approval requests and decisions to the Approval & Workflow Service.

Expense Data / Database Queries: Stores and retrieves grant expense information.

Publication Data / Database Queries: Stores and retrieves publication and citation information.

Audit Data: Records approval and status changes for accountability.

6. Justification

Reason 1 – Modularity and Maintainability

The tracker contains distinct domains such as grants, expenses, publications and approvals. Microservices isolate these responsibilities, so a change to publication indexing or co-author approval can be implemented without modifying the grant or expense services.

Reason 2 – Independent Scalability and Fault Isolation

Analytics and reporting may require heavier processing than ordinary record updates. With microservices, the analytics service can be scaled independently. A problem in one service is also less likely to bring down unrelated functions.

7. Security Advantage

Role-based access can be enforced through the User & Access Management Service so that only authorized users can access grant, publication and financial information. The Audit Ledger Service also supports tamper-evident tracking of approvals and journal-status modifications.

8. Performance Benefit

Frequently used services can be independently optimized and scaled. Grant and expense operations can remain responsive while analytics workloads are processed separately, helping the system meet the requirement for responsive access under peak usage.

9. Conclusion

Microservices Architecture is appropriate for the Faculty Research Grant & Publication Tracker because it provides clear modularity, independent scalability, fault isolation, controlled access to sensitive information and efficient handling of analytics and approval workloads.